ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2
PAPER 2 November 2023
2 hours
PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2023 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [ ].
INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) Name two structures found in a plant cell but not in an animal cell. [2]<<
>>(b) State the function of the cell membrane. [2]<<
>>(c) Describe how to prepare a slide of onion epidermis to view under a light microscope. [3]<<
>>(d) Explain why muscle cells contain many mitochondria. [3] **[10]**

2. (a) Name the enzyme in saliva and the substrate it acts on. [2]<<
>>(b) State where bile is made and where it is stored. [2]<<
>>(c) Describe two ways the ileum is adapted for absorption. [3]<<
>>(d) Explain the role of hydrochloric acid in the stomach. [3] **[10]**

3. (a) State what is transported in xylem and in phloem. [2]<<
>>(b) Define transpiration. [2]<<
>>(c) Explain how the structure of xylem is adapted to its function. [3]<<
>>(d) Suggest two environmental conditions that increase the rate of transpiration. [3] **[10]**

4. (a) Name the type of pathogen that causes malaria. [1]<<
>>(b) Name the vector of malaria. [1]<<
>>(c) Describe how malaria is transmitted from an infected person to a healthy person. [4]<<
>>(d) Suggest two methods of controlling the spread of malaria. [4] **[10]**

5. (a) State the pH of a strong acid and of a neutral solution. [2]<<
>>(b) Write a word equation for an acid reacting with a metal. [2]<<
>>(c) Write a word equation for an acid reacting with an alkali. [2]<<
>>(d) Describe a laboratory test for hydrogen gas. [2]<<
>>(e) Name the salt formed from hydrochloric acid and sodium hydroxide. [2] **[10]**

6. (a) Place potassium, copper and zinc in order of decreasing reactivity. [2]<<
>>(b) Explain why potassium is stored under oil. [2]<<
>>(c) Describe what is observed when magnesium is added to copper(II) sulfate solution. [3]<<
>>(d) State one use of aluminium and a property that makes it suitable. [3] **[10]**

7. (a) State the principle of conservation of energy. [2]<<
>>(b) A 4 kg mass is lifted through 10 m. Take g = 10 N/kg. Calculate the gain in gravitational potential energy. [3]<<
>>(c) Name the energy transfer that occurs in a battery-powered torch when it is switched on. [2]<<
>>(d) Explain how heat is transferred through the metal wall of a kettle by conduction. [3] **[10]**

8. (a) Name the three wires in a three-pin plug and state the colour of the earth wire (modern). [3]<<
>>(b) Explain the function of the fuse. [3]<<
>>(c) Explain why a metal toaster should be earthed. [2]<<
>>(d) Why must a fuse be placed in the live wire? [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).